I built a fully local RAG pipeline over five 10-K filings and measured where it breaks

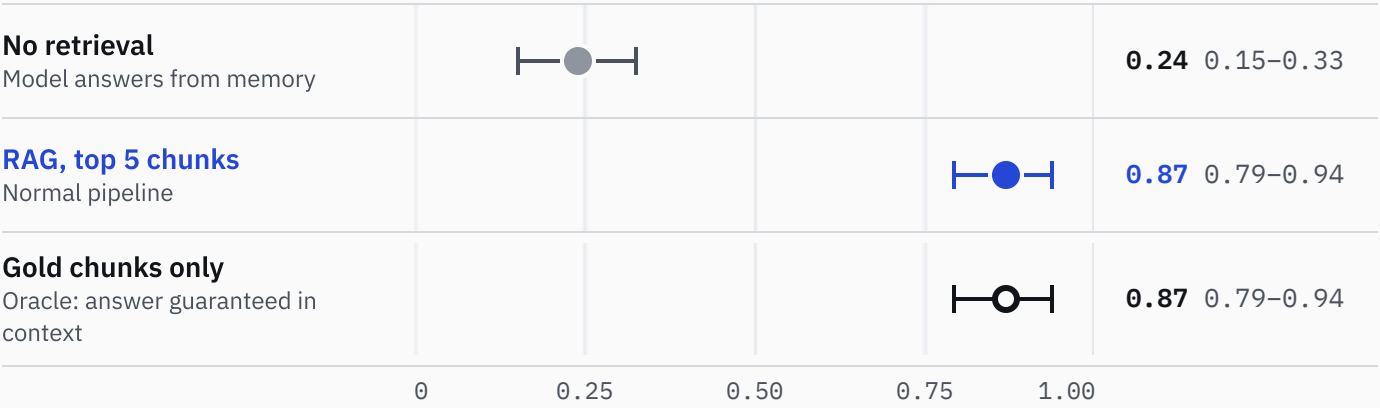
Microsoft, NVIDIA, Tesla, Lockheed Martin and BlackRock. A 4-bit Llama 3.1 8B runs on a Mac through MLX, no API calls, tested on 65 questions I checked by hand. Short version: retrieval is necessary, but it's not what limits the pipeline. Giving the model the perfect chunks scores the same.

github.com/Max-1902/local-rag-10k-eval

0.87	0.87	0.92	0.24
RAG answer score	**Gold chunks only**	**Retrieval hit@5**	**No retrieval**
95% CI 0.79–0.94	Δ vs. RAG −0.003	95% CI 0.86–0.98	95% CI 0.15–0.33

0.24 without retrieval, 0.87 with it, and still 0.87 with perfect retrieval

Answer score over all 65 questions. Dot = score, line = 95% bootstrap CI.



Why the no-retrieval row is there: these filings are newer than the model's training data, so it can't answer from memory. It scores 0.00 on every numeric question. That shows retrieval is necessary, but not how good it is. The gold-chunk row answers that question, and page 3 covers it.

Note: per company, RAG scores run from 0.74 to 0.97. With 13 questions each the intervals are about ±0.2, so differences between companies aren't meaningful and I don't compare them. Scores are conservative: a few correct answers still lose points to strict substring matching.

How it works and how I tested it

01
10-K PDF
Text pulled out with PyMuPDF

02
Heading-aware chunks
Each financial table stays whole, labeled with its heading

03
Hybrid search
Embeddings (BGE-small, Chroma) plus BM25 keyword search, merged by rank

04
Local LLM
Llama 3.1 8B, 4-bit, MLX on a Mac. Top 5 chunks in the prompt

HOW IT EVOLVED

Dense search only, then heading-aware chunking, then hybrid search. On an earlier, smaller Microsoft set, hybrid search took narrative retrieval from 0.83 to 1.00. The numeric answer score stayed at 0.56: it fixed one question and broke another. So changing the algorithm didn't move that number.

65 hand-verified questions, 13 per company

6 Narrative
Risk factors, business description, legal matters

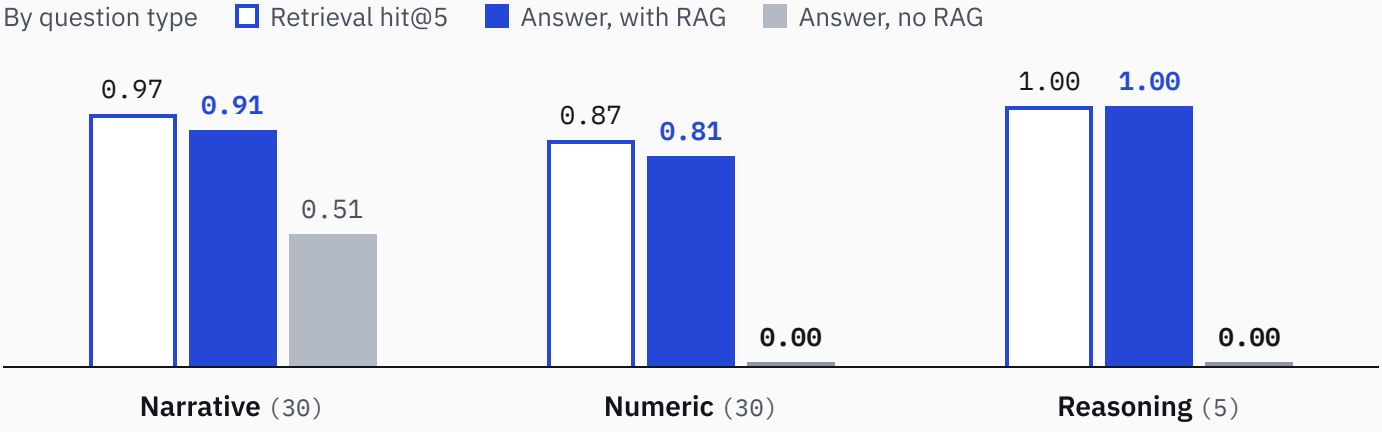
6 Numeric
One figure from a financial statement. The hard case: answers sit in dense tables

1 Reasoning
Compute a margin or share from two retrieved figures. Not printed anywhere

Every expected lookup answer was checked word for word against the PDF, and reasoning answers were computed from it. Later label fixes are logged. **Retrieval hit@5:** a top-5 chunk comes from a page that states the answer (±1 page). **Answer score:** share of expected facts that show up in the answer. **No RAG:** same model, same question, no retrieval.

Without retrieval the model can't answer a single numeric question

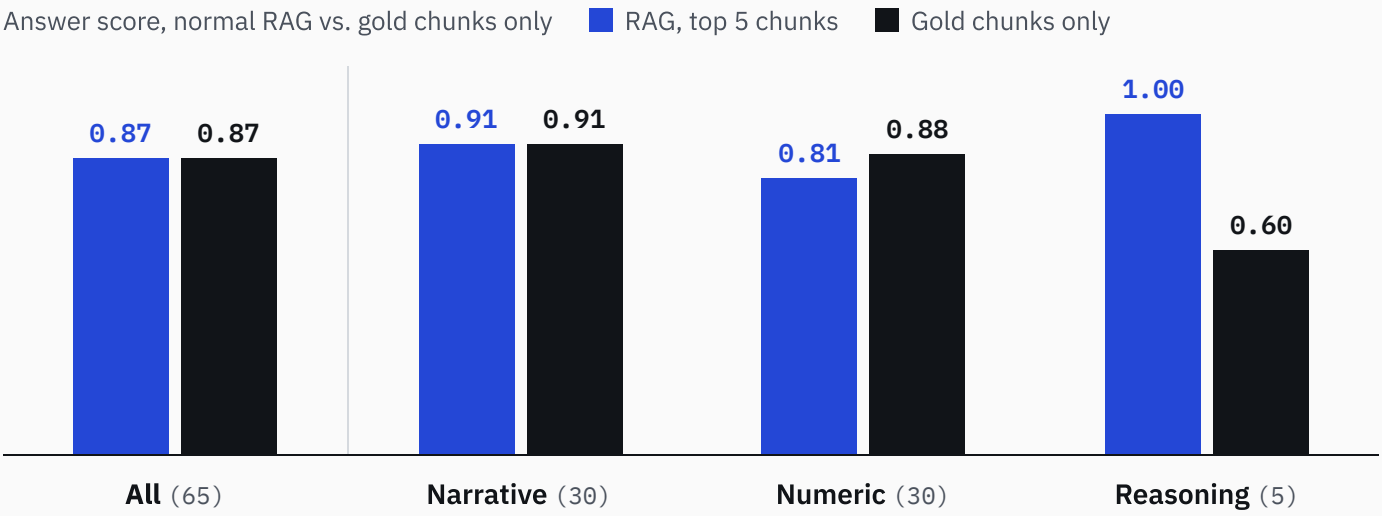
The filings are newer than its training data. This shows why retrieval is needed, not how good it is.



03 · ORACLE TEST

Perfect retrieval doesn't raise the score

At first, "the failures happen after retrieval" was something I inferred by reading cases. To measure it, I skipped retrieval and gave the model only the gold chunks, the ones that contain the answer. Same prompt, same greedy decoding, same scoring.



Paired difference, gold minus RAG: −0.003 (95% CI −0.10 to +0.10). With the answer guaranteed to be in context, the model still scores 0.87. At this sample size, retrieval isn't what limits the pipeline. Generation and scoring are.

The flat total hides movement both ways

7 better with gold context

The 2 answers retrieval truly missed, 2 over-literal refusals (given the actual cash flow statement, it answers), 1 figure it overlooked in a noisy context, 1 left-out sentence, 1 legal name that sat on only one page.

8 worse with gold context

4 picked the wrong row in a table with repeated labels, 1 lost context that sat outside the chunk (the company name), 2 missed a detail that was in the chunk, 1 arithmetic slip in the second decimal.

More context beat perfect context

BlackRock reports a GAAP table and an "As adjusted" table with identical row labels. With only the gold chunk, the model returned the As-adjusted net income, $7,736M instead of $5,553M. In normal RAG, other retrieved pages restated the GAAP figure, and that redundancy steered it to the right row.

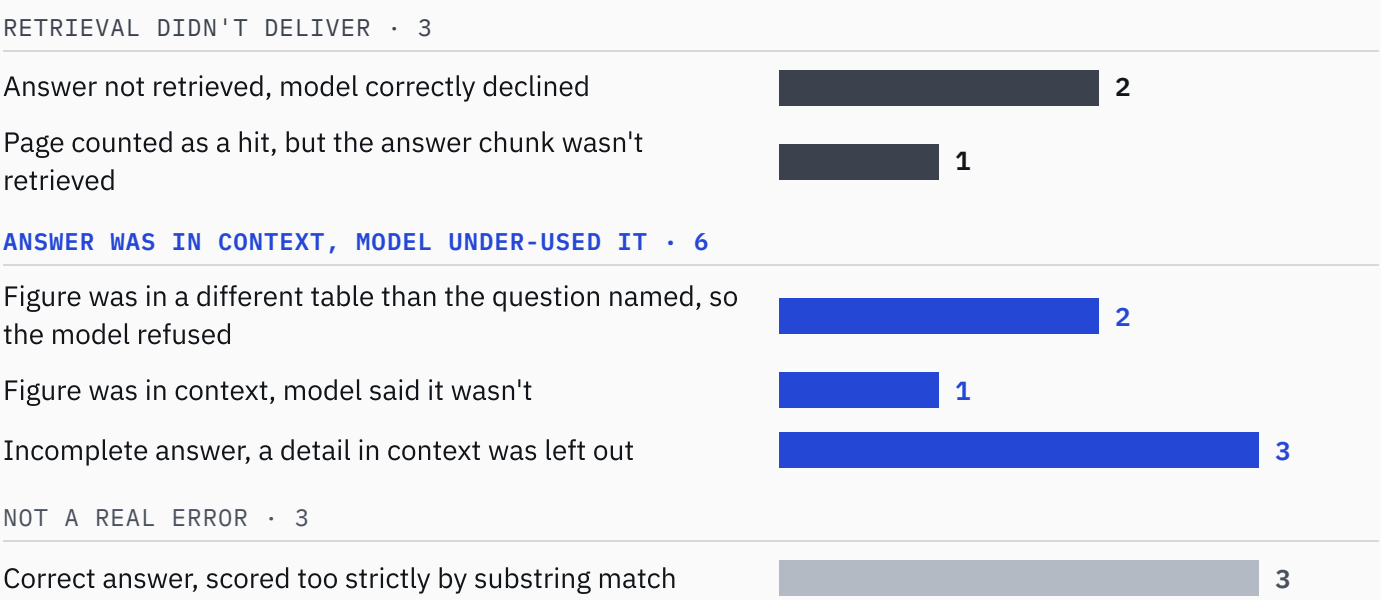
Note: this oracle uses the minimal gold context, usually one chunk. Gold chunks added on top of normal retrieval is the next run, and would separate "retrieval missed it" from "context composition" more cleanly.

What broke under normal RAG, and why

12 of the 60 lookup answers lost points, and only 3 of those were retrieval problems. In 6, the right text was in the model's context and it still left out a detail, refused too literally, or missed a figure.

Half of the lost answers had the right text in context

Why 12 of 60 narrative and numeric answers lost points, by cause



EXAMPLE · TESLA The segment note with energy revenue of $12,771M was in context. The model said the amount wasn't stated.

Page-level hit@5 is wrong in both directions

The 60 lookup questions, counted two ways



4 answers were in context even though their labeled page wasn't retrieved, because 10-Ks repeat figures. 1 page counted as a hit, but the chunk with the answer never arrived.

GROUNDING CHECK, ALL 65 ANSWERS

Every number was checked and every sentence screened against the 5 chunks the model received; flagged ones were read by hand. No invented figures or claims. One perfect score was a false positive: a vague answer passed because the expected keywords were already in the question.

Lessons and next steps

01 **Retrieval is necessary, not the bottleneck.** 0.24 to 0.87 with retrieval, and perfect retrieval adds nothing on top. What's left is generation and scoring.

02 **More context can beat perfect context.** Redundant chunks restate figures and keep the model off the wrong row. A minimal "perfect" context made 8 answers worse and 7 better.

03 **Check chunks, not pages, and audit the wins too.** Page-level hit@5 was wrong in both directions, and reading the 1.00s turned up a false positive.

04 **Next:** gold chunks added on top of normal retrieval, and the parent table header kept in every chunk so repeated row labels stay unambiguous.

05 **Then:** re-run dense vs. BM25 vs. hybrid on the current 65 questions, freeze the set, and write a held-out batch before running it.

Reasoning: 5/5 with retrieval, 3/5 with gold context only. Not robust.

Compute a margin or share from two figures in the filing

Company	Metric	With RAG	Gold only	Ground truth
MSFT	Operating margin, FY26 / FY25	46.8% / 45.6%	**46.76% / 45.59%**	46.78% / 45.62%
NVDA	Net margin	55.6%	correct	55.60%
TSLA	Energy share of revenue	13.5%	correct	13.47%
LMT	Net margin	6.68%	correct	6.69%
BLK	Net margin	23.0%	**GAAP/adj. mix-up**	22.93%

HOW I CHECKED IT

Reading every lost answer turned up three bugs in my own eval: a millions-vs-billions false negative, a "23%" vs. "23.0%" substring miss, and a wrong country count in the ground truth. My first two "hallucination" calls were wrong too, because I checked whole pages, once the wrong one, instead of the chunks the model actually saw.

LIMITATIONS

n=65 (13 per company), and I wrote both the questions and the system. Labels were adjusted after seeing results under a written rule (every change raised a score); a held-out set is planned. All 65 answers were grounding-checked, with one false positive, and a few narrative checks reuse keywords from the question. Substring scoring misses some correct paraphrases. Decoding is greedy, so reruns are identical.